Linear System — Homework 4

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Exercise 1

Determine whether or not the following matrix is nondefective.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

Solution

$$\begin{aligned} \det(A - sI) &= \begin{bmatrix} -s & 1 & 0 \\ 0 & -s & 1 \\ -1 & 1 & 1-s \end{bmatrix} \\ &= -s\Big(s(s-1)-1\Big) + (-1)\cdot 1 \\ &= -s(s^2 - s - 1) - 1 \\ &= -s^3 + s^2 + s - 1 \\ &= -(s-1)^2(s+1) \end{aligned}$$

for $s = 1$

$$A - sI = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = 1$, AM $= 2$, and GM $= 1$

for $s = -1$

$$A - sI = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = -1$, AM $= 1$, and GM $= 1$

Therefore, matrix A is defective.

Exercise 2

What is the companion matrix whose eigenvalues are -1, -2, and -3?

Solution

$$\begin{aligned} p(s) &= (s+1)(s+2)(s+3) \\ &= a_3 s^3 + a_2 s^2 + a_1 s + a_0 \\ &= s^3 + 6s^2 + 11s + 6 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \end{aligned}$$

Exercise 3

What is the real 2×2 companion matrix with eigenvalues $1 + j$, $1 - j$?

Solution

$$\begin{aligned} p(s) &= (s - (1+j))(s - (1-j)) \\ &= a_2 s^2 + a_1 s + a_0 \\ &= s^2 - 2s + 2 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -2 & 2 \end{bmatrix} \end{aligned}$$

Exercise 4

Suppose A is a real square matrix and the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -1, 2, and j, respectively. What is the response $x(t)$ of the system

$$\dot{x} = Ax$$

to the following initial conditions?

(a) $x(0) = \begin{pmatrix} 2 \\ -2 \\ 2 \\ -2 \end{pmatrix}$

Solution

Since $\mathbf{v}_1$ is an eigenvector corresponding to eigenvalue $\lambda_1 = -1$,

$$x(0) = \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = 2\mathbf{v}_1 \triangleq \mathbf{v}$$

$$A\mathbf{v} = \lambda\mathbf{v} \Leftrightarrow x(t) = e^{\lambda t}\mathbf{v} = e^{\lambda t}x(0)$$

$$\begin{aligned} x(t) &= e^{\lambda_1 t}x(0) \\ &= e^{\lambda_1 t}\mathbf{v} \\ &= e^{-t} \cdot 2\mathbf{v}_1 \\ &= \begin{bmatrix} 2e^{-t} \\ -2e^{-t} \\ 2e^{-t} \\ -2e^{-t} \end{bmatrix}. \end{aligned}$$

(b) $x(0) = \begin{pmatrix} -1 \\ -2 \\ -4 \\ -8 \end{pmatrix}$

Solution

$$x(0) = \begin{bmatrix} -1 \\ -2 \\ -4 \\ -8 \end{bmatrix} = - \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = -\mathbf{v}_2 \text{ which corresponding to eigenvalue is } 2$$

$$A\mathbf{v} = \lambda\mathbf{v} \Leftrightarrow x(t) = e^{\lambda t}x(0) = e^{\lambda t}\mathbf{v}$$

$$\begin{aligned} x(t) &= e^{\lambda t}x(0) \\ &= e^{\lambda t}\mathbf{v} \\ &= e^{\lambda t} \cdot -\mathbf{v}_2 \\ &= e^{2t}\begin{bmatrix} -1 \\ -2 \\ -4 \\ -8 \end{bmatrix} = \begin{bmatrix} -e^{2t} \\ -2e^{2t} \\ -4e^{2t} \\ -8e^{2t} \end{bmatrix} \end{aligned}$$

(c) $x(0) = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 9 \end{pmatrix}$

Solution

$$x(0) = \mathbf{v} = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 9 \end{bmatrix} = -\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = -\mathbf{v}_1 + \mathbf{v}_2$$

with linear combination

$$\begin{aligned} x(t) &= -e^{\lambda_1 t}\mathbf{v}_1 + e^{\lambda_2 t}\mathbf{v}_2 \\ &= \begin{bmatrix} -e^{-t} + e^{2t} \\ e^{-t} + 2e^{2t} \\ -e^{-t} + 4e^{2t} \\ e^{-t} + 8e^{2t} \end{bmatrix} \end{aligned}$$

(d) $x(0) = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$

Solution

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix} = \mathbf{u}_3 + j\mathbf{p}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + j\begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow \mathbf{u}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{p}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda_3 = \alpha + j\omega = j \Rightarrow \alpha = 0, \quad \omega = 1$$

$$\begin{aligned} x(0) &= \mathbf{u} + \mathbf{p} \\ x(t) &= x_u(t) + x_p(t) \\ &= e^{\alpha t}\Big(\cos(\omega t) \cdot \mathbf{u} - \sin(\omega t) \cdot \mathbf{p}\Big) + e^{\alpha t}\Big(\sin(\omega t) \cdot \mathbf{u} + \cos(\omega t) \cdot \mathbf{p}\Big) \end{aligned}$$

with $\alpha = 0, \quad \omega = 1$

$$\begin{aligned} x(t) &= \mathbf{u} \cdot (\cos t + \sin t) + \mathbf{p} \cdot (\cos t - \sin t) \\ &= \begin{bmatrix} \cos t + \sin t \\ \cos t - \sin t \\ -\cos t - \sin t \\ -\cos t + \sin t \end{bmatrix} \end{aligned}$$

Exercise 5

Consider a discrete-time LTI system described by

$$x(k+1) = Ax(k).$$

(a) Suppose that the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -2 and 3, respectively. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}?$$

Solution

$$x(0) = \frac{1}{2}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right) = \mathbf{v} = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$$

$$\begin{aligned} x(k) &= A^k \left(\frac{1}{2}\mathbf{v}_1 + \frac{1}{2}\mathbf{v}_2 \right) \\ &= \frac{1}{2}A^k\mathbf{v}_1 + \frac{1}{2}A^k\mathbf{v}_2 \\ &= \frac{1}{2}\lambda_1^k\mathbf{v}_1 + \frac{1}{2}\lambda_2^k\mathbf{v}_2 \\ &= \frac{1}{2}(-2)^k \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \frac{1}{2}(3)^k \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} (-2)^k + 3^k \\ -(-2)^k + 3^k \\ (-2)^k + 3^k \\ -(-2)^k + 3^k \end{bmatrix} \end{aligned}$$

(b) Suppose A is a real matrix and the vector

$$\begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

is an eigenvector of A corresponding to the eigenvalue $2+3j$. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}?$$

Solution

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + j \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \mathbf{u} + j\mathbf{p}$$

$$x(0) = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} = \operatorname{Re}(\mathbf{v}) = \mathbf{u}$$

Let $r = \sqrt{13}$ and $\theta = \arctan(3/2)$, so that $2+3j = r(\cos\theta + j\sin\theta)$. Since A is real,

$$\begin{aligned} x(k) &= A^k x(0) = A^k \operatorname{Re}(\mathbf{v}) \\ &= \operatorname{Re}(A^k\mathbf{v}) = \operatorname{Re}\big((2+3j)^k\mathbf{v}\big) \\ &= r^k \operatorname{Re}\big[(\cos(k\theta) + j\sin(k\theta))\mathbf{v}\big] \\ &= r^k \operatorname{Re}\big[(\cos(k\theta) + j\sin(k\theta))(\mathbf{u}+j\mathbf{p})\big] \\ &= r^k\big(\cos(k\theta)\mathbf{u} - \sin(k\theta)\mathbf{p}\big). \end{aligned}$$

$$x(k) = \sqrt{13}^k \left(\cos(k\arctan(3/2)) \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} - \sin(k\arctan(3/2)) \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} \sqrt{13}^k \cos(k\arctan(3/2)) \\ -\sqrt{13}^k \sin(k\arctan(3/2)) \\ -\sqrt{13}^k \cos(k\arctan(3/2)) \\ \sqrt{13}^k \sin(k\arctan(3/2)) \end{bmatrix}$$

Exercise 6

Consider a 5×5 matrix

$$A = vw^T$$

where v and w are 5×1 matrices.

(a) What is the rank of A?

Solution

for $v, w \neq 0$

$$A = vw^T = \begin{bmatrix} w_1 v & w_2 v & w_3 v & w_4 v & w_5 v \end{bmatrix}$$

Hence,

$$\text{range}(A) = \text{span}\{v\}$$

thus,

$$\text{rank}\, A = \dim(\text{range}(A)) = 1$$

but if v or w is 0, $\text{rank}\, A = 0$

(b) What is the nullity of A?

Solution

if v or $w \neq 0$

$$\text{null}\, A = \dim A - \text{rank}\, A = 5 - 1 = 4$$

else

$$\text{null}\, A = 5$$

(c) What are the eigenvalues of A if $w^T v \neq 0$?

Solution

$$A = vw^T \Rightarrow Av = v(w^T v)_{1\times 1} = (w^T v)v = \lambda v$$

Hence, the eigenvalue is $w^T v$

Since the nullity of A is 4, there are four linearly independent eigenvectors corresponding to eigenvalue 0. Therefore, the eigenvalues are

$$w^T v, \quad 0, \quad 0, \quad 0, \quad 0$$

For Exercise 7 and 8, please use Matlab/Python to illustrate the time history plot. Please don't plot y, θ_1, θ_2 in the same plot, separate them!

Exercise 7

Recall the 2 pendulum cart system.

$$\begin{aligned}(m_0+m_1+m_2)\ddot{y} - m_1 l_1 \cos\theta_1\,\ddot{\theta}_1 - m_2 l_2 \cos\theta_2\,\ddot{\theta}_2 \\ + m_1 l_1 \sin\theta_1\,\dot{\theta}_1^2 + m_2 l_2 \sin\theta_2\,\dot{\theta}_2^2 = u \\ - m_1 l_1 \cos\theta_1\,\ddot{y} + m_1 l_1^2 \ddot{\theta}_1 + m_1 l_1 g \sin\theta_1 = 0 \\ - m_2 l_2 \cos\theta_2\,\ddot{y} + m_2 l_2^2 \ddot{\theta}_2 + m_2 l_2 g \sin\theta_2 = 0\end{aligned}$$

Consider the equilibrium configurations defined by $u^e = 0$ and

$$E1: \qquad (y^e, \theta_1^e, \theta_2^e) = (0,0,0)$$

and

$$E2: \qquad (y^e, \theta_1^e, \theta_2^e) = (0,\pi,\pi).$$

Consider state space representations of the linearizations corresponding to the following combinations of parameters and equilibrium conditions:

	Parameter	Equilibrium
$L1$	$P1$	$E1$
$L2$	$P1$	$E2$
$L3$	$P2$	$E1$
$L4$	$P2$	$E2$
$L5$	$P3$	$E1$
$L6$	$P3$	$E2$
$L7$	$P4$	$E1$
$L8$	$P4$	$E2$

(a) Determine the eigenvalues of all the ***linearized systems*** L1–L8.

Solution

E1: $(y^e, \theta_1^e, \theta_2^e) = (0,0,0)$

Construction of the state-space model. Near the equilibrium

$$\sin\theta_i \approx \theta_i, \quad \cos\theta_i \approx 1$$

The terms $\sin\theta_i\,\dot{\theta}_i^2$ are neglected because they are higher-order terms in the perturbations.

The linearized equations become

$$\begin{cases}(m_0+m_1+m_2)\ddot{y} - m_1 l_1 \ddot{\theta}_1 - m_2 l_2 \ddot{\theta}_2 & = u \\ -m_1 l_1 \ddot{y} + m_1 l_1^2 \ddot{\theta}_1 + m_1 l_1 g \theta_1 & = 0 \\ -m_2 l_2 \ddot{y} + m_2 l_2^2 \ddot{\theta}_2 + m_2 l_2 g \theta_2 & = 0\end{cases}$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} y \\ \theta_1 \\ \theta_2 \\ \dot{y} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \Rightarrow \dot{x} = \begin{bmatrix} \dot{y} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}$$

obtained with rewritten equation

$$\ddot{\theta}_1 = \frac{\ddot{y} - g\theta_1}{l_1}, \text{ and } \ddot{\theta}_2 = \frac{\ddot{y} - g\theta_2}{l_2}$$

Substituting into the first equation gives

$$\begin{aligned} u &= (m_0 + m_1 + m_2)\ddot{y} - m_1(\ddot{y} - g\theta_1) - m_2(\ddot{y} - g\theta_2) \\ &= m_0\ddot{y} + m_1 g\theta_1 + m_2 g\theta_2. \end{aligned}$$

$$\ddot{y} = \frac{-m_1 g\theta_1 - m_2 g\theta_2 + u}{m_0}$$

Hence,

$$\begin{aligned} \ddot{\theta}_1 &= \frac{\ddot{y} - g\theta_1}{l_1} = \frac{-(m_0 + m_1)g\theta_1 - m_2 g\theta_2 + u}{m_0 l_1}, \\ \ddot{\theta}_2 &= \frac{\ddot{y} - g\theta_2}{l_2} = \frac{-m_1 g\theta_1 - (m_0 + m_2)g\theta_2 + u}{m_0 l_2}. \end{aligned}$$

$$\begin{aligned} \dot{x} = \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} &= \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \frac{-m_1 g}{m_0} x_2 - \frac{m_2 g}{m_0} x_3 + \frac{1}{m_0} u \\ \frac{-(m_0+m_1)g}{m_0 l_1} x_2 - \frac{m_2 g}{m_0 l_1} x_3 + \frac{1}{m_0 l_1} u \\ \frac{-m_1 g}{m_0 l_2} x_2 - \frac{(m_0+m_2)g}{m_0 l_2} x_3 + \frac{1}{m_0 l_2} u \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-m_1 g}{m_0} & \frac{-m_2 g}{m_0} & 0 & 0 & 0 \\ 0 & \frac{-(m_0+m_1)g}{m_0 l_1} & \frac{-m_2 g}{m_0 l_1} & 0 & 0 & 0 \\ 0 & \frac{-m_1 g}{m_0 l_2} & \frac{-(m_0+m_2)g}{m_0 l_2} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ \frac{1}{m_0 l_1} \\ \frac{1}{m_0 l_2} \end{bmatrix} u \\ &= Ax + Bu \end{aligned}$$

L1 (P1) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 1, 1)$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-m_1 g}{m_0} & \frac{-m_2 g}{m_0} & 0 & 0 & 0 \\ 0 & \frac{-(m_0+m_1)g}{m_0 l_1} & \frac{-m_2 g}{m_0 l_1} & 0 & 0 & 0 \\ 0 & \frac{-m_1 g}{m_0 l_2} & \frac{-(m_0+m_2)g}{m_0 l_2} & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-1}{2} & \frac{-1}{2} & 0 & 0 & 0 \\ 0 & \frac{-3}{2} & \frac{-1}{2} & 0 & 0 & 0 \\ 0 & \frac{-1}{2} & \frac{-3}{2} & 0 & 0 & 0 \end{bmatrix}$$

L3 (P2) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.99, 1)$

L5 (P3) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 0.5, 1, 1, 1)$

L7 (P4) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.5, 1)$

E2: $(y^e, \theta_1^e, \theta_2^e) = (0, \pi, \pi)$

Construction of the state-space model.

L2 (P1) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 1, 1)$

L4 (P2) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.99, 1)$

L6 (P3) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 0.5, 1, 1, 1)$

L8 (P4) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.5, 1)$

(b) Compare the behavior of the nonlinear system with that of the linearized system for cases L7 and L8. Illustrate your results using different initial condition with time histories of y, θ_1, and θ_2. Plesae write paragraph to analysis the comparison between nonlinear and linearized system.

Initial conditions for case L7: ± 0 deg, ± 1 deg, ± 10 deg, ± 90 deg, ± 180 deg.

Initial conditions for case L8: ± 1 deg.

Solution

Exercise 8

This exercise refers to linearizations L7 and L8 of the two pendulum cart system.

(a) For L7 choose an initial state for the linearized system which results in a periodic solution for the linearized system.

Solution

(b) For L8 choose an initial state for the linearized system which results in a solution which asymptotically decays to zero for the linearized system.

Solution

(c) For L8 choose an initial state for the linearized system which results in a solution whose magnitude grows exponentially for the linearized system.

Solution

In each case, simulate both the linearized system and the nonlinear system with initial conditions corresponding to your chosen initial state for the linearized system.

MATLAB Source Code

The following listings reproduce the source files used to generate the figures.

Main script

```
clear; clc; close all;
% Homework 4, Exercises 7(b) and 8: nonlinear/linear comparisons.
% Run this script, with cart_pendulum_model.m in the same folder.
rootDir = fileparts(mfilename('fullpath'));
addpath(rootDir);
outputDir = fullfile(rootDir, 'figures');
if ~exist(outputDir, 'dir'), mkdir(outputDir); end

% Confirmed interpretation: use the Homework 1 parameter sets.
% Columns: m0, m1, m2, l1, l2, g.
P = [2 1 1   1 1    1;
     2 1 1   1 0.99 1;
     2 1 0.5 1 1    1;
     2 1 1   1 0.5  1];
models = cell(8, 1);
eigenvalues = zeros(6, 8);
for k = 1:8
    models{k} = cart_pendulum_model(P(ceil(k/2), :), 2-mod(k, 2));
    eigenvalues(:, k) = eig(models{k}.A);
    % Independent finite-difference check of the implemented Jacobian.
    J = zeros(6); h = 1e-6;
    for j = 1:6
        ej = zeros(6, 1); ej(j) = h;
        J(:, j) = (models{k}.rhs(0, models{k}.xe+ej) ...
                   -models{k}.rhs(0, models{k}.xe-ej))/(2*h);
    end
    assert(norm(J-models{k}.A, 'inf') < 1e-7);
end

options = odeset('RelTol', 1e-10, 'AbsTol', 1e-12, 'MaxStep', 0.02);
results = struct('id', {}, 'p', {}, 'A', {}, 'xe', {}, 'z0', {}, ...
    'x0', {}, 't', {}, 'zLinear', {}, 'xNonlinear', {}, ...
    'zNonlinear', {}, 'mode', {});

%% Exercise 7(b): opposite angular deviations from the equilibrium.
angles = [0 1 10 90 180];
for alpha = angles
    z0 = [0; deg2rad(-alpha); deg2rad(alpha); 0; 0; 0];
    id = sprintf('ex7_L7_%03ddeg', alpha);
    results(end+1) = simulate_case(models{7}, z0, 30, id, options);
end
z0 = [0; deg2rad(-1); deg2rad(1); 0; 0; 0];
results(end+1) = simulate_case(models{8}, z0, 5, ...
    'ex7_L8_001deg', options);

%% Exercise 8: select modes numerically, leaving written derivations blank.
% 8(a): real part of an oscillatory mode, scaled to 1 degree peak angle.
[V7, D7] = eig(models{7}.A);
lambda7 = diag(D7);
candidates = find(imag(lambda7) > 1e-8);
[~, j] = min(imag(lambda7(candidates)));
j = candidates(j);
v = V7(:, j);
[~, pivot] = max(abs(v(1:3)));
v = v*exp(-1i*angle(v(pivot)));
z0 = scale_angles(real(v), 1);
period = 2*pi/imag(lambda7(j));
r = simulate_case(models{7}, z0, 4*period, 'ex8a_L7_periodic', options);
assert(norm(expm(models{7}.A*period)*z0-z0) < 1e-9);
r.mode = lambda7(j); results(end+1) = r;

% 8(b,c): negative/positive real modes of L8.
[V8, D8] = eig(models{8}.A);
```

```
lambda8 = diag(D8);
for signChoice = [-1 1]
    candidates = find(signChoice*real(lambda8) > 1e-8 ...
                      & abs(imag(lambda8)) < 1e-8);
    [~, j] = min(abs(real(lambda8(candidates))));
    j = candidates(j);
    v = real(V8(:, j));
    [~, pivot] = max(abs(v(2:3))); pivot = pivot+1;
    if v(pivot) < 0, v = -v; end
    z0 = scale_angles(v, 1);
    if signChoice < 0
        id = 'ex8b_L8_decay'; tf = 10;
    else
        id = 'ex8c_L8_growth'; tf = 5;
    end
    % Evaluate a pure real mode analytically to avoid numerical contamination
    % by the unstable modes when illustrating asymptotic linear decay.
    r = simulate_case(models{8}, z0, tf, id, options, real(lambda8(j)));
    r.mode = lambda8(j); results(end+1) = r;
end

%% Export separate plots, report fragments, and a reloadable workspace.
for k = 1:numel(results)
    export_case(results(k), outputDir, rootDir);
end
fid = fopen(fullfile(rootDir, 'simulation-ex7.tex'), 'w');
assert(fid ~= -1);
fprintf(fid, '\\clearpage\n\\subsection*{MATLAB time histories}\n');
fprintf(fid, ['Parameters P1--P4 follow Homework 1. ' ...
    'The two pendulum lengths are $l_1$ and $l_2$; for P4, ' ...
    '$l_1=1$, $l_2=0.5$.\n\n']);
fprintf(fid, ['State order: $x=[y,\\theta_1,\\theta_2,\\dot y,' ...
    '\\dot\\theta_1,\\dot\\theta_2]^T$. Both models start from the same ' ...
    'physical state $x(0)=x^e+z(0)$, with $u=0$. ' ...
    'Plots show deviations $z=x-x^e$; angles are in degrees and are not wrapped.\n\n']);
fprintf(fid, ['For Exercise 7(b), $z(0)=[0,-\\alpha,+\\alpha,0,0,0]^T$ ' ...
    '(angular entries converted from degrees to radians). ' ...
    'L7 uses $\\alpha=0,1,10,90,180$ deg and L8 uses $\\alpha=1$ deg. ' ...
    'L7 is simulated for 30 s and L8 for 5 s.\n\n']);
fprintf(fid, ['Nonlinear simulation: \\texttt{ode45}, relative tolerance ' ...
    '$10^{-10}$, absolute tolerance $10^{-12}$, maximum step 0.02 s. ' ...
    'Linear responses use the matrix exponential.\n']);
for k = 1:6
    fprintf(fid, '\\input{simulation-%s.tex}\n', results(k).id);
end
fclose(fid);
save(fullfile(rootDir, 'hw4_workspace.mat'), ...
    'P', 'models', 'eigenvalues', 'results', 'options');
fprintf('Finished: %d cases, %d separate plots.\n', numel(results), 3*numel(results));
fprintf('Workspace saved to %s\n', fullfile(rootDir, 'hw4_workspace.mat'));

function z = scale_angles(v, degrees)
z = v*(deg2rad(degrees)/max(abs(v(2:3))));
end

function r = simulate_case(model, z0, tf, id, options, mode)
t = linspace(0, tf, 1501).';
[t, xNonlinear] = ode45(model.rhs, t, model.xe+z0, options);
zLinear = zeros(numel(t), 6);
if nargin == 6
    assert(norm(model.A*z0-mode*z0) < 1e-9);
    zLinear = exp(t*mode)*z0.';
else
    for k = 1:numel(t)
        zLinear(k, :) = (expm(model.A*t(k))*z0).';
    end
end
r = struct('id', id, 'p', model.p, 'A', model.A, 'xe', model.xe, ...
    'z0', z0, 'x0', model.xe+z0, 't', t, 'zLinear', zLinear, ...
    'xNonlinear', xNonlinear, 'zNonlinear', xNonlinear-model.xe.', ...
    'mode', NaN);
assert(all(isfinite(xNonlinear(:))) && all(isfinite(zLinear(:))));
assert(norm(r.zNonlinear(1, :).'-z0) < 1e-12);
```

```
end

function export_case(r, outputDir, rootDir)
names = {'y', 'theta1', 'theta2'};
labels = {'\Delta y (m)', '\Delta\theta_1 (deg)', '\Delta\theta_2 (deg)'};
captionNames = {'$\Delta y$', '$\Delta\theta_1$', '$\Delta\theta_2$'};
fid = fopen(fullfile(rootDir, ['simulation-' r.id '.tex']), 'w');
assert(fid ~= -1);
fprintf(fid, '\\clearpage\n\\subsection*{\\texttt{%s}}\n', strrep(r.id, '_', '\_'));
fprintf(fid, 'Initial deviation $z(0)$ (radians for angles, SI units otherwise):\n');
fprintf(fid, '\\begin{center}\\scriptsize\\texttt{[');
fprintf(fid, '%.7g, ', r.z0(1:5)); fprintf(fid, '%.7g]}\\end{center}\n', r.z0(6));
fprintf(fid, '$x^e=[0,%.7g,%.7g,0,0,0]^T$, $x(0)=x^e+z(0)$.\n', r.xe(2:3));
if startsWith(r.id, 'ex8')
    fprintf(fid, ['The initial state is selected numerically from an eigenmode; ' ...
        'see the source code. Linear states are deviations from equilibrium.\n']);
end
for j = 1:3
    scale = 1; if j > 1, scale = 180/pi; end
    fig = figure('Visible', 'off', 'Color', 'w', ...
        'Position', [100 100 1000 270]);
    ax = axes(fig, 'Position', [0.12 0.24 0.84 0.60]);
    plot(ax, r.t, scale*r.zNonlinear(:, j), '-', ...
        'Color', [0 0.35 0.65], 'LineWidth', 1.4); hold(ax, 'on');
    plot(ax, r.t, scale*r.zLinear(:, j), '--', ...
        'Color', [0.85 0.25 0.12], 'LineWidth', 1.2);
    grid(ax, 'on'); xlabel(ax, 'Time (s)'); ylabel(ax, labels{j});
    title(ax, strrep(r.id, '_', ' '));
    legend(ax, 'Nonlinear', 'Linearized', 'Location', 'best');
    set(ax, 'FontSize', 11, 'Color', 'w', 'XColor', [0.1 0.1 0.1], ...
        'YColor', [0.1 0.1 0.1]);
    exportgraphics(fig, fullfile(outputDir, [r.id '_' names{j} '.png']), ...
        'Resolution', 180, 'BackgroundColor', 'white');
    savefig(fig, fullfile(outputDir, [r.id '_' names{j} '.fig']));
    close(fig);
    fprintf(fid, '\\begin{figure}[!ht]\n\\centering\n');
    fprintf(fid, '\\includegraphics[width=0.94\\linewidth]{figures/%s_%s.png}\n', r.id, names{j});
    fprintf(fid, '\\caption{%s: nonlinear and linearized time histories.}\n\\end{figure}\n', captionNames{j
    });
end
fclose(fid);
end
```

Model function

```
function model = cart_pendulum_model(p, equilibrium)
% Parameters: [m0 m1 m2 l1 l2 g]. State: [y th1 th2 yd w1 w2].
% The linear state is the deviation z = x - xe; input u = 0.
m0 = p(1); m1 = p(2); m2 = p(3);
l1 = p(4); l2 = p(5); g = p(6);
if equilibrium == 1
    xe = zeros(6, 1);
    c = 1;
else
    xe = [0; pi; pi; 0; 0; 0];
    c = -1;
end
M = [m0+m1+m2, -m1*l1*c, -m2*l2*c;
     -m1*l1*c, m1*l1^2, 0;
     -m2*l2*c, 0, m2*l2^2];
K = diag([0, m1*l1*g*c, m2*l2*g*c]);
model.A = [zeros(3), eye(3); -(M\K), zeros(3)];
model.B = [zeros(3, 1); M\[1; 0; 0]];
model.xe = xe;
model.p = p;
model.rhs = @(t, x) nonlinear_rhs(t, x, p);
end

function dx = nonlinear_rhs(~, x, p)
m0 = p(1); m1 = p(2); m2 = p(3);
l1 = p(4); l2 = p(5); g = p(6);
s1 = sin(x(2)); s2 = sin(x(3));
c1 = cos(x(2)); c2 = cos(x(3));
% Preserve exact equilibria at integer multiples of pi in floating point.
if abs(s1) < 8*eps, s1 = 0; end
if abs(s2) < 8*eps, s2 = 0; end
ydd = (-m1*g*c1*s1 - m2*g*c2*s2 ...
       -m1*l1*s1*x(5)^2 - m2*l2*s2*x(6)^2) ...
      / (m0 + m1*s1^2 + m2*s2^2);
dx = [x(4:6); ydd; (c1*ydd-g*s1)/l1; (c2*ydd-g*s2)/l2];
end
```